

Activity

- 1 A CCTV image Contains Salt and pepper noise due to transmission errors

Scenario

CCTV image Contains Salt and pepper noise due to transmission errors

Technique used

Median Filter

Reason

Removes Salt and pepper noise effectively while preserving image edges.

- 2 Satellite image

Scenario

Satellite image is Corrupted by Gaussian noise from Sensors

Technique used

Gaussian Filter

Reason

Smooths Gaussian noise by averaging nearby pixels values

- 3 Medical X-ray image

Scenario

Medical X-ray image has random noise but fine edge must be preserved

Technique used

Bilateral Filter

Reason

Reduces Noise preserving edges

4 Scanned document

Scenario

Scanned document has black and white dots scattered randomly.

Technique use

Median Filter

Reason

Eliminates isolated black and white dots without blurring text.

5 Low-light image

Scenario

Low-light image shows heavy grain without clear boundaries.

Technique use

Gaussian filter

Reason

Reduces grainy noise and produces a smoother image.

6 Foggy road image

Scenario

Foggy road image has very low contrast.

Technique use

Histogram Equalization

Reason

Improves contrast, making the road and objects more visible.

7 Underwater image

Scenario

Underwater image appears dull and washed out.

Reason

Contrast stretching

Reason

Enhances brightness and contrast to improve image appearance.

8 Chest X-ray

Scenario

Chest X-ray shows poor visibility of internal structures.

Reason

Adaptive Histogram Equalization (CLAHE)

Reason

Enhances local contrast and reveals internal structures clearly.

9 Night-time traffic

Scenario

Night-time traffic image lacks intensity variation.

Reason

Histogram Equalization

Reason

Increases intensity variation and improves overall visibility.

10 Grayscale image uses

Scenario

Grayscale image uses only a narrow intensity

Reason

Contrast Stretching

Reason

Expands the intensity range, improving image contrast and clarity.